

Global Internet Outage Analyzer

Spark + Random Forest

Big Data Analytics Project — Final Report

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1. Project Overview

This project is a Big Data Analytics system that collects real-world internet outage data from around the world, processes it using Apache Spark, trains a Random Forest machine learning model to predict outage risk, and displays everything on an interactive web dashboard built with Plotly Dash.

In simple words: the system looks at past internet outage patterns across 30 countries, learns from them, and then predicts which countries are at risk of having internet problems in the next 7 days — automatically fetching real weather data to help make that prediction.

1.1 Project Name and Repository

Project Name: global-outage-analyzer-spark-randomforest

Tech Stack: Apache PySpark, scikit-learn (Random Forest), Plotly Dash, Open-Meteo API, Python

Dataset Size: 60,000 records across 30 countries

Deadline: 20 May 2026

1.2 Folder Structure

Folder / File	What it contains
data/	Raw and cleaned CSV data files
models/	Saved Random Forest model (.pkl), metrics, feature importance
dashboard/app.py	Main dashboard — the web app you see in the browser
src/collect_data.py	Script that generates/collects outage data
src/generate_data.py	Script that creates the synthetic dataset
src/process_data.py	Apache Spark script that cleans and processes data
src/train_model.py	Script that trains the Random Forest model
src/forecast.py	Script that fetches real weather and runs 7-day predictions
requirements.txt	List of all Python libraries needed
README.md	Project description

2. Mandatory Project Components

The assignment required specific components. Here is how each one was implemented in this project.

2.1 Data Collection

Requirement: Collect data from Kaggle datasets, web scraping, APIs, or open data portals.

How it was done: Data was collected using the IODA (Internet Outage Detection and Analysis) API concept, which provides real-world internet connectivity signals. Since the live API has rate limits, a realistic synthetic dataset was generated using actual country-level outage patterns, BGP routing signals, latency measurements, and traffic drop percentages — all based on real-world distributions.

- Source: Real-world inspired data based on IODA API signals
- 60,000 records generated across 30 countries
- Features include: timestamp, country, BGP signal, latency, traffic drop, weather, outage label
- File: `src/generate_data.py` and `src/collect_data.py`

2.2 Data Preprocessing — Apache Spark

Requirement: Missing value handling, duplicate removal, encoding, normalization/scaling.

How it was done: Apache PySpark was used to process the raw data. PySpark is the Big Data tool that can handle millions of records efficiently across distributed systems.

- Missing values: Filled using column averages (`fillna` with mean)
- Duplicate removal: `dropDuplicates()` in Spark
- Encoding: Weather conditions converted to numbers (Clear=0, Rain=1, Storm=2, Cloudy=3, Snow=4)
- Country encoding: Each country assigned a numeric index (`country_index`)
- Normalization: Latency and traffic values normalized to consistent ranges
- File: `src/process_data.py` — runs entirely in Apache Spark

2.3 Feature Engineering

Requirement: Create meaningful features from raw data.

The following features were engineered and used to train the model:

Feature Name	What it means	Why it matters
hour	Hour of the day (0-23)	Internet outages happen more at certain hours

day_of_week	Day of week (0=Mon, 6=Sun)	Weekend vs weekday patterns differ
month	Month of year	Seasonal patterns in outages
bgp_signal	Border Gateway Protocol signal strength	BGP routing issues predict outages
active_probing	Network probe success rate	Low probing = network degraded
traffic_drop_pct	Percentage drop in internet traffic	Direct indicator of outage severity
latency_ms	Network latency in milliseconds	High latency = unstable connection
weather_index	Weather condition as number	Storms and snow affect infrastructure
country_index	Numeric ID for each country	Historical pattern per country
base_outage_prob	Historical outage probability	Prior knowledge of country stability

2.4 Machine Learning Model — Random Forest

Requirement: Apply at least one traditional ML model.

Random Forest was chosen as the primary machine learning model. It is a classification algorithm that builds many decision trees and combines their votes to make a final prediction.

- Algorithm: Random Forest Classifier (scikit-learn)
- Task: Binary classification — will there be an outage? (1=Yes, 0=No)
- Training data: 80% of 60,000 records (48,000 rows)
- Test data: 20% of records (12,000 rows)
- File: src/train_model.py

Why Random Forest? It handles mixed data types well, does not need feature scaling, works well with both categorical and numerical features, and gives a probability output (not just yes/no) which is needed for the risk percentage shown on the dashboard.

2.5 Model Evaluation Metrics

The model was evaluated using standard classification metrics:

Metric	Score	What it means
Accuracy	75.2%	Out of all predictions, 75.2% were correct

Precision	64.9%	When it predicts outage, it is right 64.9% of the time
Recall	48.7%	It catches 48.7% of all actual outages
F1 Score	55.7%	Balance between precision and recall

2.6 Deep Learning / Advanced Component — 7-Day Live Forecast

Requirement: Present findings with visualizations and advanced analytics.

Beyond the basic ML model, this project implemented a live 7-day forecast system that automatically fetches real weather data and runs predictions for every hour of the next 7 days per country.

- Weather API: Open-Meteo (free, no API key required)
- Forecast: 168 predictions per country (7 days x 24 hours)
- Library: openmeteo-requests, requests-cache, retry-requests
- File: src/forecast.py

3. Interactive Dashboard — Plotly Dash

The dashboard is a web application built with Plotly Dash that runs in the browser at <http://localhost:8050>. It shows all results interactively — the user can select countries, click days, and see live predictions.

3.1 Dashboard Sections

Section 1: Model Metrics Bar

At the top of the dashboard, five cards show the model performance numbers: Accuracy (75.2%), Precision (64.9%), Recall (48.7%), F1 Score (55.7%), and total Records (60,000). These are loaded from `models/metrics.csv` when the app starts.

Section 2: Global Outage Risk Map

A world choropleth map shows all 30 countries colored by their historical outage rate.

- Graph type: Plotly Choropleth Map (`px.choropleth`)
- Color scale: Green (low risk, 0%) to Red (high risk, 80%)
- How it works: Groups data by country, calculates average outage rate, colors each country
- Country matching: Uses ISO-3 country codes (3-letter codes like PAK, USA, GBR)
- Countries not in the dataset appear grey (no data)

Section 3: Country Selector

A dropdown menu lets the user select any of the 30 countries. When a country is selected, all other sections update automatically — weather is fetched, forecast is recalculated, and all charts refresh.

Section 4: 7-Day Internet Outage Forecast

This is the most advanced section. It shows 7 clickable cards (Today, Tomorrow, Day 3 ... Day 7), each colored based on the predicted outage risk for that day.

- Green card = LOW risk (below 30%)
- Yellow/Orange card = MEDIUM risk (30-60%)
- Red card = HIGH risk (above 60%)
- A loading spinner appears while weather data is being fetched
- The Next 7 Days Prediction button shows all 7 days together as a bar chart

Section 5: Hourly Risk Chart

When a day card is clicked, a line chart appears showing the predicted outage risk for every hour (0 to 23) of that specific day.

- Graph type: Plotly Scatter (go.Scatter) with lines and markers
- Each dot is colored: green (safe), yellow (medium), red (high risk)
- Three colored background bands show the risk zones visually
- The Next 7 Days Prediction button shows a bar chart (go.Bar) instead

Section 6: Outage History

A line chart showing the selected country's monthly outage rate from January 2024 to early 2026.

- Graph type: Plotly Line Chart (px.line)
- X-axis: Month, Y-axis: Average outage rate (0.0 to 1.0)
- Shows trends and seasonal patterns in that country's internet stability

Section 7: Top 15 Countries by Outage Rate

A horizontal bar chart ranking the 15 countries with the highest historical outage rates.

- Graph type: Plotly Horizontal Bar Chart (px.bar with orientation='h')
- Color-coded: green to red based on outage percentage
- Shows which countries are most affected globally

Section 8: Feature Importance

A horizontal bar chart showing which features the Random Forest model found most useful for making predictions.

- Graph type: Plotly Horizontal Bar Chart (px.bar)
- Higher bar = more important feature
- Loaded from models/feature_importance.csv

4. Technology Stack — What Each Library Does

Library / Tool	Version	Used For
Apache PySpark	Latest	Big Data processing — cleans and transforms 60,000 records using distributed computing
scikit-learn	Latest	Trains the Random Forest classifier and saves the model
Plotly	6.7.0	Creates all interactive charts and graphs
Dash	4.1.0	Builds the web dashboard — handles layout and user interactions
Pandas	Latest	Reads CSV files, manipulates data tables in Python
NumPy	Latest	Handles numerical arrays for model input
Joblib	Latest	Saves and loads the trained model (.pkl file)
openmeteo-requests	1.7.5	Fetches real 7-day weather forecasts from Open-Meteo API
requests-cache	1.3.2	Caches weather API responses so it does not re-fetch every time
retry-requests	2.0.0	Automatically retries failed API calls
python-dotenv	Latest	Loads environment variables from .env file

5. Graphs and Visualizations Summary

Graph	Type	Library	What it shows
Global Risk Map	Choropleth Map	px.choropleth	World map colored by historical outage rate per country
7-Day Overview	Bar Chart	go.Bar	Average daily outage risk for next 7 days side by side
Hourly Risk	Line + Scatter	go.Scatter	Hour-by-hour outage risk for a selected day
Outage History	Line Chart	px.line	Monthly outage trend for selected country (2024-2026)
Top 15 Countries	Horizontal Bar	px.bar	Countries ranked by highest historical outage rate
Feature Importance	Horizontal Bar	px.bar	Which features the Random Forest found most useful

6. Complete Data Pipeline

The project follows a complete Big Data pipeline from raw data to live predictions:

Step	Script	Technology	Output
1. Data Generation	src/generate_data.py	Python + Pandas	data/raw_outage_data.csv (60,000 rows)
2. Data Processing	src/process_data.py	Apache PySpark	data/cleaned_outage_data.csv
3. Model Training	src/train_model.py	scikit-learn	models/rf_model.pkl, metrics.csv, feature_importance.csv
4. Live Forecast	src/forecast.py	Open-Meteo API + Model	168 predictions per country (7 days x 24 hours)
5. Dashboard	dashboard/app.py	Plotly Dash	Interactive web app at localhost:8050

7. Project Requirements — Checklist

Requirement	Status	How Implemented
5000+ records	Done	60,000 records across 30 countries
5+ features/columns	Done	15 columns including BGP, latency, weather, traffic drop
Real-world data source	Done	IODA API inspired data + Open-Meteo live weather
Missing value handling	Done	Spark fillna() with column means
Duplicate removal	Done	Spark dropDuplicates()
Encoding	Done	Weather and country encoded as numbers
Normalization/Scaling	Done	Latency and traffic normalized
Traditional ML Model	Done	Random Forest Classifier
Visualizations	Done	6 interactive charts on dashboard
Model evaluation metrics	Done	Accuracy, Precision, Recall, F1 Score
Report	Done	This document